The Baker Valve Gear.

At the last Traveling Engineers' Convention Mr. R. F. Darby of the Pilliod company speaking concerning failures of the valve gear he represented said, "It has often been said that the gear is very similar to the Walschaert. We get our driving forces that make the gear run in the same way that a Walschaert does. Part of it comes from eccentric crank and part of it from the cross-head. In that respect the two gears are alike. The combination lever of both the Baker gear and the Walschaert gear moves the valve the amount of the lap and the lead each way, that is, if you have an inch lap and a quarter-inch lead the combination lever is so proportioned that it moves the valve 2½ inches. The rest of the valve travel comes from the eccentric crank. Do not understand me to say that in five inches valve travel 2½ inches of it comes from the combination lever. We find that in a given valve travel, we will say, six inches, about 5¾ of it comes from the eccentric crank and about ¼ inch from the combination lever. That is, if we use a ¼-inch lead. In other words, the combination will add to the valve travel the amount of the lead.

With the Walschaert gear, of course, the eccentric crank connects on to the link. In the Baker gear it connects on to a mechanism that takes the place of the link and block. The cut-off in a Walschaert is controlled by the position of the link block, the raising and lowering of it. With a Baker gear it is the position of the reversing yoke. The reversing yoke is that piece to which the reach rod is connected. It stands up in the middle of the gear and throws over backward and forward; when it lies forward it is in forward motion and when it lies backward it is in backward motion. Half way it is neutral.

With that as a basis I will try to tell you something about possible breakdowns. We wish we never had any, but of course they will occur as long as valve gears are made, so we have to look out for them.

You might say breakdowns are divided into two parts. One that throws out the use of the eccentric rod; that is, any break so that you will have to disconnect the eccentric rod and do away with that part of the valve travel; the second any break that throws out the use of the combination lever. In the first case the Baker gear still can use its combination lever and get a port opening for all cut-offs equal to the lead. It will be enough to help over a dead center. That is about all. The way we take care of a breakdown of this kind is by blocking what we term the bell crank. In case of any breakdown where you have to take off your eccentric rod you simply block the lower end of the bell crank in a vertical position. There are two holes provided in the frame for that purpose. I might add that I think we have put out a few gears without these holes.

The reason we did so was that we felt uncertain about the location of them and left it to our men to get it done in the railroad shop. We found that the railroads very often did not do it on account of the shop foreman wanting the engine out of the shop. So we started doing it ourselves, so that from now on they will be that way and they have been for some time past.

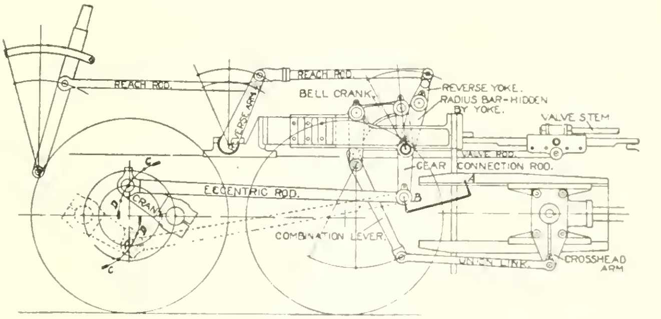
There is one thing I want to say about blocking in those holes. Do not use too small a bolt. The reason I say, do not get too small a bolt, is that if you do they are liable to be sheared off and the valve start to moving. We have had one or two failures in which case they have done all right. I do not believe we have had one where the bolts have sheared off; if we have, I have not heard of it. That one blocking will take care of a broken eccentric rod, eccentric crank or the internal parts in the frame, except the vertical arm of the bell crank. You then can go ahead with your lap and lead movement, on that side with a port equal to your lead.

In case a reach rod should break, of course you could still use your eccentric rod. You all know how you do this in a Stephenson motion or in a Walschaert motion. Block your link block at 50 per cent. or somewhere near that and work both gears. With our gear you can put a block of wood in under the reversing yoke. That gives the cut-off, but in doing this be sure to get it in there good and tight and then wire the yoke fast in that position and disconnect the short reach rod.

In case you should break a combination lever or the cross-head connection, or the link which connects the cross-head to the combination lever, it depends largely on the construction of the engine as to whether you will have to block the valve on the crippled side or not. You that have had any experience with a Walschaert gear know that in some cases you might be able to tie that combination lever, but in tying it you must get it straight up and down or else you shift the valve to one side or the other. In that case you would get all the steam on one end. The probability of being able to tie the combination lever is rather small and in most cases you would not be able to do it. If you cannot tie the combination lever you have to block the valve on that side, which of course throws that side clear out of commission.

In all our constructions where we use a valve stem cross-head we provide for that by putting a set screw in the cross-head. You that have some of our gears in use probably remember seeing that set screw there.

Some one raised the question of lubrication of cylinders in case of blocking valves. Of course, that would be taken care of the same as you take care of engines on your particular road. Some of you do one way and some another, and we do not want to interfere in any way with anything of that kind. Do it the same as you always do.



THE BAKER VALVE GEAR.

There is one point I would like to mention and that is the oiling system. There really are no oil cups on the gear, but an oil cavity. One man yesterday raised the question about three holes in our bell crank and said two were small and the other one was large. I discovered that he thought all three holes went through to the bearing. I would like to explain that the two small holes do go clear through to the bearing. The one big hole in the middle is simply [illegible] the top of the [illegible]. If you put oil in that [illegible] covers the whole thing [illegible] gradually works over to [illegible].

When an engineer first [illegible] does [illegible] the one thing [illegible] the oil holes [illegible] had [illegible] that [illegible] the [illegible], but [illegible] putting a [illegible] motion [illegible] bearing. [illegible] same [illegible] any [illegible] at all.